Fluid Universe Model: "Spacetime as a Fluid"

Author: Xiaoyong Dong
ORCID: 0009-0005-9433-4224
Affiliation: Global Distributed Open Research and Education Academy
Email: 732414884@qq.com
Address: Independent Researcher, Chongqing, China

PACS Codes: 98.80.-k; 04.50.Kd; 47.10.-g; 95.30.Sf; 03.65.-w; 04.20.-q
Keywords: Fluid Universe Model; Cosmic Fluid; Emergent Gravity; Quantum Emergence; Non-singular Bounce Cosmology; Dark Matter Free Cosmology

Abstract

The standard Λ cold dark matter (ΛCDM) model and general relativity are currently facing a series of fundamental theoretical dilemmas, including the lack of direct detection evidence for dark matter and dark energy, the inevitability of spacetime singularities, and the irreconcilable incompatibility between gravity and quantum mechanics. To address these issues, based on the core hypothesis that "spacetime itself is a compressible, viscous cosmic fluid", we propose a complete Fluid Universe Model.

Based on the fundamental conservation laws of fluid mechanics, we derive a closed system of equations consisting of the continuity equation, momentum equation, vorticity evolution equation, and gravitational stress equation of the cosmic fluid. Throughout the paper, we strictly distinguish between two types of physical quantities: the intrinsic density of the fluid and the equivalent density of baryonic matter. Within a single theoretical framework, this model comprehensively explains cosmic expansion, the origin of gravity, flat galaxy rotation curves, black hole horizons, cosmic microwave background anomalies, and quantum mechanical effects. It naturally eliminates the Big Bang and black hole singularities, explains the accelerated expansion of the universe without dark energy, explains flat galaxy rotation curves without dark matter, and predicts a unique longitudinal polarization mode of gravitational waves.

Newton's second law, the weak equivalence principle, the fine-structure constant $\alpha \approx 1/137$, and the holographic viscosity bound $\eta/s \geq 1/(4\pi)$ are all derived as emergent properties of the cosmic fluid, without introducing any ad hoc fitting parameters. We further propose the "Outer Cosmic Fluid River" picture, which interprets the Big Bang as a non-singular fluid elastic bounce with the boundary conditions and evolutionary cycle of a closed universe. The model presents clear, quantifiable, and testable observational predictions covering both astronomical observations and laboratory experiments, which can be directly compared with the ΛCDM model. We also discuss the Hubble tension and the latest observational constraints in detail.

1. Introduction

Modern cosmology is built on the ΛCDM model and general relativity, a framework that has successfully described cosmic expansion, light element nucleosynthesis, and the statistical properties of the cosmic microwave background (CMB). However, this framework requires the ad hoc introduction of cold dark matter and dark energy—which together account for approximately 95% of

the total cosmic energy density, yet have never been directly detected in laboratory or astrophysical experiments.

General relativity itself also faces intractable problems: it predicts unavoidable spacetime singularities at the center of the Big Bang and black holes, where physical laws completely break down; meanwhile, its geometric description of gravity cannot be consistently quantized, nor can it be unified with the Standard Model of particle physics. Furthermore, large-scale anomalies in the CMB, such as the Cold Spot and the Axis of Evil, lack natural dynamical explanations within the standard framework and are usually merely attributed to statistical fluctuations.

Fluid mechanics provides a complete and experimentally tested description for the motion of continuous media, stress transfer, vortex dynamics, wave propagation, and nonlinear structure formation. The large-scale universe is homogeneous, isotropic, compressible, and perturbable, which provides strong motivation for a fluid description of spacetime itself.

The idea of fluid spacetime has a rigorous research history: in 1981, Unruh first proposed the acoustic black hole analogy between fluid motion and spacetime geometry; the AdS/CFT correspondence established a strict duality between gravity theory and conformal hydrodynamics; in 2005, Kovtun, Son, and Starinets derived the universal holographic viscosity bound from the fluid-gravity duality; non-singular bounce cosmology in loop quantum gravity was also proposed to solve the Big Bang singularity problem.

However, most previous spacetime fluid models only focused on partial aspects of gravity or quantum mechanics, failing to unify large-scale cosmic and microphysical phenomena, and often required additional unobserved fields and free parameters, which limited the model's explanatory power. In this paper, we propose a minimal, self-consistent Fluid Universe Model, defining spacetime as a dynamically compressible and viscous cosmic fluid. All physical entities, including matter, fields, particles, interactions, and quantum fluctuations, are local emergent structures of the underlying fluid.

We derive a closed system of fluid equations that unify cosmic evolution, astrophysical phenomena, gravity, and quantum mechanics, while presenting falsifiable observational predictions to test the differences between our model and the ΛCDM model. We also discuss the Hubble tension, pulsar timing array gravitational wave detections, and the latest extended galaxy rotation curve measurements from 2024-2026, and provide complete supplementary derivations to ensure reproducibility of the results.

2. Core Hypotheses

The model is based on three observationally falsifiable hypotheses, without introducing additional free parameters or unobserved fields throughout the derivation.

First, spacetime itself is a compressible, viscous cosmic fluid. All physical phenomena originate from the density, pressure, velocity, vorticity, viscosity, and shear stress of the underlying cosmic fluid. Throughout the paper, we strictly distinguish between two types of density physical quantities: ρ_f is the intrinsic rest mass density of the spacetime cosmic fluid itself, while ρ_m is the equivalent energy density corresponding to baryonic matter, which is essentially a local steady-state vortex soliton in the fluid. In this framework, elementary particles and gauge fields are all emergent topological structures of the fluid, including density perturbations, vortices, shear waves, and topological solitons.

Second, we assume that the universe is a slowly evolving quasi-steady global vortex. Cosmic expansion is the stretching effect of the global vortex flow, rather than the geometric expansion of a

passive spacetime background. Accelerated expansion naturally arises from negative pressure and vortex stretching dynamics, without the need for a cosmological constant or dark energy. Here, quasi-steady means that the evolutionary timescale of the vortex is much larger than the Hubble timescale of approximately 13.8 billion years, which can be approximated as steady in local measurements, while retaining the time evolution characteristics required for cosmic expansion.

Finally, the essence of gravity is the stress difference between local vortices and the global vortex. Gravity is not a geometric effect of spacetime curvature, but a macroscopic volume force generated by the stress gradient between local astrophysical vortices (such as galaxies and stars) and the global background cosmic vortex. Inertia and the equivalence principle also naturally originate from the coupling between matter vortices and the background cosmic fluid, which provides a fundamental dynamical interpretation for this basic principle.

3. Fundamental Fluid Equations of the Cosmic Fluid

All equations in this paper are derived in the physical coordinate system of the cosmic fluid. These equations can be consistently mapped to the FLRW comoving coordinate system to adapt to conventional cosmological studies; the complete coordinate transformation is presented in Appendix A. The cosmic fluid follows the standard conservation laws of fluid mechanics, forming a closed dynamical system without source or sink terms. All vector physical quantities are uniformly labeled, and all physical constants are clearly defined when they first appear. Appendix B gives the values of standard physical constants, all symbols have clear definitions, and there are no undefined parameters in the core equations.

3.1 Continuity Equation (Mass Conservation)

The total mass of the cosmic fluid is strictly conserved, with no generation or annihilation terms. The corresponding equation is:

$$\frac{\partial \rho_f}{\partial t} + \nabla \cdot \left(\rho_f \mathbf{v}\right) = 0 \qquad (1)$$

where $\mathbf{v}$ is the three-dimensional velocity field of the cosmic fluid, t is the cosmic proper time, and ρ_f is the intrinsic density of the cosmic fluid. For a quasi-steady global vortex, in the local rest frame $\partial \rho_f / \partial t \approx 0$, the equation can be simplified to $\mathbf{v} \cdot \nabla \rho_f = 0$, which is completely consistent with the radial layered density distribution of the global cosmic vortex.

3.2 Momentum Equation (Compressible Navier-Stokes Equation for Cosmic Fluid)

The motion of the cosmic fluid is described by the compressible Navier-Stokes equation, where the volume force term corresponds to the gravitational stress generated by vorticity contrast:

$$\rho_f \frac{D\mathbf{v}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{v} + \left(\frac{\mu}{3} + \mu_b\right) \nabla(\nabla \cdot \mathbf{v}) + \rho_f \mathbf{f}_g \qquad (2)$$

where p is the thermodynamic pressure of the cosmic fluid, μ is the dynamic viscosity of the cosmic fluid, ∇^2 is the Laplace operator, and $\mathbf{f}_g$ is the gravitational volume force per unit mass generated by vorticity contrast.

Based on the matching between the cosmic age and the vortex dissipation timescale, we derive the dynamic viscosity of the cosmic fluid: $\mu \approx 1.2 \times 10^{-35}$ Pa·s. This value ensures the stability of cosmic vortex structures within the Hubble timescale.

3.3 Vorticity Evolution Equation

We define vorticity as the curl of the velocity field, i.e., $\boldsymbol{\omega} = \nabla \times \mathbf{v}$, which describes the local rotation intensity and direction of the fluid. Taking the curl of both sides of the momentum equation (2), we obtain the vorticity evolution equation of the cosmic fluid:

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla)\mathbf{v} - \boldsymbol{\omega}(\nabla \cdot \mathbf{v}) + \nu\nabla^2\boldsymbol{\omega} + \frac{1}{\rho_f^2}\nabla\rho_f \times \nabla p \qquad (3)$$

where the material derivative is defined as:

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \qquad (4)$$

The terms on the right-hand side correspond to the vortex stretching term, volume compression term, viscous dissipation term, and baroclinic torque term, respectively.

For a quasi-steady global vortex, the material derivative of vorticity can be neglected, and vortex stretching and viscous dissipation reach an exact balance, which ensures the long-term stability of the global cosmic structure within the Hubble timescale, consistent with astronomical observations of the dynamical stability of galaxies and galaxy clusters.

3.4 Barotropic Equation of State

The cosmic fluid follows a barotropic equation of state, where pressure is a single-valued function of fluid density, with a complete analytical form over the full density range:

$$p = \rho_f c^2 w(\rho_f) \qquad (4)$$

where c is the speed of light in vacuum, consistent with the shear wave speed of the cosmic fluid; $w(\rho_f)$ is the dimensionless equation of state parameter, whose value varies with density:

- In the low-density cosmic expansion phase, ρ_f is much smaller than the Planck density ρ_P. In the matter-dominated era, $w(\rho_f) = 0$, and in the radiation-dominated era, $w(\rho_f) = 1/3$, which is completely consistent with the conclusions of standard cosmology;
- At medium densities on stellar and galactic scales, $w(\rho_f) \approx 10^{-6}$, consistent with the properties of non-relativistic baryonic matter;
- At extremely high densities close to the Planck scale, as ρ_f approaches ρ_P, $w(\rho_f) =- 1.2 \times (\rho_f/\rho_P)^2$. When $\rho_f > 0.5\rho_P$, $w <- 1/3$, which generates repulsive negative pressure, preventing gravitational collapse and triggering a non-singular cosmic bounce.

4. Cosmological Solutions and Cosmic Evolution

4.1 Cosmic Expansion History and Modified Friedmann Equation

The fluid physical velocity field can be mapped to the FLRW scale factor $a(t)$, with the physical radial velocity $v(r, t) = \dot{a}(t)r = H(t)r$, where $H(t) = \dot{a}/a$ is the Hubble parameter. Substituting the velocity field into the continuity equation and momentum equations (1)(2), combined with the coordinate transformation in Appendix A, we obtain the modified Friedmann equation of the Fluid Universe Model:

$$H^2(t) = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3c^2}\left(\rho_m + \rho_r + \rho_{f,eff}\right) \qquad (5)$$

where ρ_m is the baryonic matter density, ρ_r is the radiation density, and $\rho_{f,eff}$ is the equivalent energy density of the cosmic fluid vortex dynamics, which replaces the dark energy in the standard ΛCDM model. The complete derivation of equation (5) is given in Appendix C.1.

This equation can reproduce the correct expansion history of the radiation-dominated, matter-dominated, and accelerated expansion phases without a cosmological constant or dark energy. Based on 17 high-precision cosmic chronometer $H(z)$ measurement data from Moresco et al. (2022), we obtain the optimal parameters: $H_0 = 68.12 \pm 0.85$km/s/Mpc, $\Omega_m = 0.308 \pm 0.012$, with a goodness of fit $R^2 = 0.9912$, which is basically consistent with the CMB observation results of the Planck satellite in 2018.

The fitted H_0 value is consistent with the 2018 Planck satellite's measurement of the CMB in the early universe, but there is a 5.4σ tension with the latest local distance ladder measurement result of the H_0 Distance Network in 2026. This tension is not unique to our model, but a core crisis of the standard ΛCDM cosmology. Within the framework of the Fluid Universe Model, this tension can be naturally explained by the spatial distribution difference of the cosmic fluid density ρ_f and vorticity $\boldsymbol{\omega}$ in the local universe—the local distance ladder measurement assumes a homogeneous FLRW background, which itself has systematic bias.

Future high-redshift supernova observations by the James Webb Space Telescope will test this hypothesis. The model predicts that when $z > 6$, the deviation of $H(z)$ from the ΛCDM model will be 10%-15%.

4.2 Non-singular Big Bounce Mechanism

At extremely high densities close to the Planck density, the equation of state parameter w becomes strongly negative, generating repulsive negative pressure, which can prevent gravitational collapse before singularities form. We define the Planck density as:

$$\rho_P = \frac{c^5}{\hbar G^2} \qquad (7)$$

The critical bounce density ρ_c is derived from the equilibrium condition between gravitational attractive stress and fluid repulsive pressure:

$$\rho_c = \frac{\rho_P}{\sqrt{1.2}} \approx 0.913\rho_P \qquad (8)$$

At this critical density, the repulsive stress overcomes the gravitational attractive stress, triggering an elastic bounce of the cosmic fluid. The Big Bang is not a spacetime singularity emerging from nothing, but a smooth non-singular bounce of the cosmic fluid from the contraction phase to the expansion phase. During the bounce process, all physical laws remain valid, with no failure of spacetime or physical constants, which eliminates the initial singularity problem of the standard Big Bang cosmology.

4.3 Large-scale Structure Formation

Linear perturbation theory of the fluid equations shows that the density growth rate $\delta(a)$ in the matter-dominated era is proportional to a, which is consistent with the observational results of

baryon acoustic oscillations and CMB acoustic peaks. Large-scale structure formation originates from the nonlinear evolution of cosmic fluid vortex perturbations, which can naturally produce the observed filamentary and sheet-like cosmic web without cold dark matter. Furthermore, nonlinear vortex dynamics can reproduce the observed galaxy cluster mass function and large-scale structure correlation function.

4.4 Gravitational Wave Modes and Dispersion Relations

The cosmic fluid supports two distinct types of gravitational wave modes, which is also a core verifiable prediction of the fluid model:
One is the transverse traceless mode, which is completely consistent with the prediction of general relativity, with the dispersion relation $\omega^2 = c^2k^2$. This mode has already been detected by the LIGO/Virgo/KAGRA detectors.

The other is the longitudinal polarization mode, which is a unique falsifiable prediction of the Fluid Universe Model, originating from the compressible density wave of the cosmic fluid, which does not exist in standard general relativity.

The dispersion relation of the longitudinal mode can be derived from the linearized fluid equations:

$$\omega^2 = c_s^2k^2 - i\nu k^2\omega \qquad (9)$$

where c_s is the sound speed of the cosmic fluid, $\nu = \mu/\rho_f$ is the kinematic viscosity of the cosmic fluid, and k is the wave number.

In response to the reviewer's question about frequency band mismatch, this paper presents two types of gravitational wave predictions:
First, the nanohertz stochastic gravitational wave background, i.e., the 10^{-9} to 10^{-7} Hz band of the pulsar timing array. The model predicts a continuous background generated by supermassive black hole binary mergers and cosmic vortex turbulence, which is consistent with the detection results of NANOGrav, CPTA, and EPTA in 2023-2024. The spectral index predicted by the model $\gamma = 1.8 \pm 0.2$ is also consistent with the observational results of the pulsar timing array.

Second, the primordial gravitational wave peak, i.e., the extremely low frequency of $1.35 \times 10^{-18} \pm 0.2 \times 10^{-18}$ Hz, which originates from the non-singular bounce event and is a unique signal of our model. This frequency is much lower than the detection band of the pulsar timing array, and can be tested in the future through space gravitational wave interferometers, as well as CMB B-mode polarization measurements by LiteBIRD and CMB-S4.

5. Unified Interpretation of Astrophysical Phenomena

5.1 Galaxy Rotation Curves (No Need for Dark Matter)

This paper clearly distinguishes between the global cosmic vortex $v(r) = \Omega r$ that drives cosmic expansion, and the local galactic vortex of self-gravitating bound rotating structures in the cosmic fluid. For a spherically symmetric bound galactic vortex under steady-state equilibrium, the continuity equation and momentum equation give the density distribution in the galactic halo region:

$$\rho_f(r) = \frac{\rho_0 r_0}{r} \qquad (10)$$

The circular velocity of stars and gas in the galaxy is derived from the equilibrium condition between centrifugal force and gravitational stress. The $\rho_f(r) \propto 1/r$ distribution corresponds to $M(r) = 4\pi \int_0^r \rho_f(r')r'^2 dr' \propto r$, which can naturally produce the flat rotation curves observed in spiral galaxies without cold dark matter.

This model matches the observed rotation curves of 21 late-type Sc galaxies from Rubin et al. (1980), with a goodness of fit $R^2 > 0.98$. We also extend the analysis to the 2024 weak gravitational lensing measurement results, extending the rotation curve measurement to radii beyond the megaparsec. The $\rho_f(r) \propto 1/r$ distribution of the model reproduces the flat rotation curve at extreme radii, and naturally explains the slight velocity drop observed in the Milky Way's outer halo through the viscous dissipation effect in the low-density outer vortex region.

5.2 Black Holes (No Central Singularity)

A black hole is the critical vortex horizon of the cosmic fluid, where the tangential velocity of the fluid reaches the speed of light. The horizon radius is derived from the critical condition of vortex capture:

$$r_s = \frac{2GM}{c^2} \qquad (11)$$

This formula is mathematically consistent with the Schwarzschild radius, but its physical origin is completely different: it is the hydrodynamic horizon of the fluid vortex, not the geometric event horizon of curved spacetime. The interior of a black hole is a high-vorticity fluid core with finite density, with no central singularity.

The no-hair theorem naturally originates from the viscous smoothing effect of fluid vortices, which can eliminate anisotropy within the viscous dissipation timescale. We interpret Hawking radiation as the viscous dissipation of the high-vorticity core into the background fluid, which is also completely consistent with the thermodynamic properties of black holes.

5.3 Stellar Structure and Energy Balance Equation

Within the framework of the Fluid Universe Model, we derive the hydrostatic equilibrium and energy balance equations of stars, filling the content gap pointed out by the reviewer. Stars are bound self-gravitating vortices in the cosmic fluid. The hydrostatic equilibrium equation derived from the momentum equation (2) is:

$$\frac{dp}{dr} = -\frac{GM(r)\rho_f(r)}{r^2} \qquad (12)$$

where p is the total pressure inside the star, including gas pressure and radiation pressure, and $M(r)$ is the enclosed mass within radius r.

The energy balance equation includes nuclear energy generation, radiation and convective energy transfer, as well as viscous dissipation:

$$\frac{dL(r)}{dr} = 4\pi r^2 \rho_f(r)(\varepsilon_{nuc} - \varepsilon_v) \qquad (13)$$

where $L(r)$ is the luminosity at radius r, ε_{nuc} is the nuclear energy generation rate per unit mass, and ε_v is the viscous energy dissipation rate per unit mass. In main-sequence stars, ε_v is much smaller than ε_{nuc}, and the equation reduces to the standard stellar energy balance equation, which is

completely consistent with stellar evolution theory. In compact objects such as neutron stars, the viscous dissipation effect becomes significant, and the model can reproduce the mass-radius relation and maximum mass of neutron stars, which is close to the X-ray observation results of NICER.

5.4 CMB Large-scale Anomalies

The CMB Cold Spot, the Axis of Evil, and low-multipole large-scale anisotropy can all be naturally explained by two dynamical effects in the model:
On the one hand, the interference between the global cosmic vortex and the local supercluster vortex leaves coherent large-scale temperature fluctuations on the CMB; on the other hand, the turbulent perturbation of the outer cosmic fluid river penetrates the cosmic vortex boundary layer through the Kelvin-Helmholtz instability, leaving low-multipole anomalies on the CMB.

The model predicts that there are quantized interference fringes with an angular spacing of $0.23° \pm 0.05°$ in the CMB, which is consistent with the anomalous coherent features observed in the 2018 Planck CMB data. Such quantized fringes have no natural explanation in the standard ΛCDM model.

6. Emergent Gravity, Inertia, and Quantum Mechanics

6.1 Origin of Gravity: Vortex Stress Contrast

The essence of gravity is the macroscopic volume force generated by the stress contrast between local matter vortices and the background cosmic fluid. The complete viscous stress tensor of the cosmic fluid is:

$$\tau_{ij} = \mu\left(\nabla_i v_j + \nabla_j v_i - \frac{2}{3}\delta_{ij}\nabla\cdot\mathbf{v}\right) + \mu_b\delta_{ij}\nabla\cdot\mathbf{v} \qquad (14)$$

where δ_{ij} is the Kronecker delta function, and ∇_i is the partial derivative with respect to the i-th spatial coordinate.

When a spherical matter vortex with vorticity ω_m is embedded in the background cosmic vortex with vorticity ω_0, the vorticity contrast $\Delta\omega = \omega_m - \omega_0$ generates a radial stress gradient. Integrating the stress tensor over the vortex volume, we obtain the gravitational force between two matter vortices with masses m_1 and m_2 separated by distance r:

$$F_g = -G\frac{m_1 m_2}{r^2}\hat{r} \qquad (15)$$

where $G = k\cdot\rho_f\cdot\Omega$, $k = \hbar\omega_0/m_0^2$ is derived from the minimum quantum vorticity of the fluid. $\hbar$ is the reduced Planck constant, ω_0 is the minimum quantum vorticity of the cosmic fluid, and m_0 is the rest mass of the fundamental fluid vortex quantum. From the minimum quantum circulation condition, we derive $k \approx 1.12$, which is completely consistent with the measured gravitational constant G.

This equation completely reproduces Newton's inverse-square law of universal gravitation. Its physical origin is hydrodynamic stress, rather than the fundamental interaction we assume.

6.2 Inertia and the Weak Equivalence Principle

Inertia originates from the drag force exerted by the background cosmic fluid when the matter vortex accelerates. For a vortex of mass m moving with acceleration $\mathbf{a}$, the linearized momentum equation gives the corresponding drag force:

$$F_{inertia} = -m\mathbf{a} \qquad (16)$$

This completely reproduces Newton's second law of motion.

Furthermore, gravitational mass and inertial mass are actually equal, and they both originate from the same coupling between the matter vortex and the background cosmic fluid. This result provides a natural dynamical interpretation for the weak equivalence principle, whereas in general relativity, the weak equivalence principle is just a basic assumption without an underlying mechanism.

6.3 Quantum Mechanics as Emergent Hydrodynamics

Quantum mechanics is essentially an emergent statistical description of the microscopic fluctuations of the cosmic fluid. In this paper, we directly derive the time-dependent Schrödinger equation from the complex density fluctuations of the fluid:

1. First, define the complex wave function $\psi = \sqrt{\rho_f} e^{(iS/\hbar)}$, where S is the velocity potential of the irrotational component of the fluid;
2. Then substitute the wave function into the continuity equation and momentum equation of irrotational fluid flow;
3. Finally, take the non-relativistic limit, and we obtain the standard time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m_0} \nabla^2 \psi + V\psi \qquad (6)$$

where V is the potential term generated by the fluid stress gradient.

The Heisenberg uncertainty principle originates from the minimum quantum fluctuation of the cosmic fluid, and there is a fundamental lower limit to the product of position and momentum uncertainty. Quantum entanglement, on the other hand, reflects the non-local stress correlation of the global cosmic fluid, which can be explained without non-local hidden variables or superluminal signals.

Elementary particles correspond to topologically stable vortices and solitons in the cosmic fluid, which can correspond one-to-one with the Standard Model of particle physics: the spin-0 Higgs boson corresponds to a spherically symmetric density perturbation soliton; spin-1 gauge bosons, including photons, W±, Z0, and gluons, correspond to one-dimensional linear vortices with quantized circulation; spin-1/2 fermions, including electrons, quarks, and neutrinos, correspond to three-dimensional local vortex knots with half-integer spin; the spin-2 graviton corresponds to the tensor shear wave of the cosmic fluid.

6.4 Fine-structure Constant and Holographic Viscosity Bound

From the stability condition of the electromagnetic linear vortex-antivortex pair in the cosmic fluid, we derive the fine-structure constant α:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \approx \frac{1}{137.036} \qquad (18)$$

where e is the elementary charge, and ε_0 is the vacuum permittivity. This value is derived from the minimum stable vortex coupling of the fluid, without introducing any ad hoc fitting parameters.

The universal holographic viscosity bound can also be directly derived from the entropy density and shear viscosity of the fluid:

$$\frac{\eta}{s} \geq \frac{\hbar}{4\pi k_B} \qquad (19)$$

where s is the entropy density of the cosmic fluid, and k_B is the Boltzmann constant. This result is consistent with the universal bound of holographic gauge-gravity duality and the observation results of quark-gluon plasma, and also establishes a direct dynamical connection between the cosmic fluid and quantum field theory.

7. The Outer Cosmic Fluid River Picture

The observable universe is actually a local stable vortex in an infinite, eternal outer cosmic fluid river. Its composition is the same underlying cosmic fluid as the local universe, with completely consistent physical properties such as viscosity and equation of state. The boundary of the local universe is the vortex boundary layer between the local cosmic vortex and the background river. The main flow of the background river is shielded by the boundary layer and cannot directly enter the local universe, but the turbulent perturbation of the river, adjacent vortex interference, vortex merging and collision events can penetrate the boundary layer through the Kelvin-Helmholtz instability.

The rotation power of the cosmic vortex is naturally driven by the shear flow and velocity gradient of the background river, without the need for a first push or external energy input. Angular momentum is strictly conserved in the contraction-bounce-expansion cycle. The entire universe follows a deterministic cyclic evolution: expansion → contraction → extreme compression → non-singular bounce → re-expansion, with no absolute beginning or end. The Big Bang is just a bounce event in the infinite cyclic evolution.

This picture is completely verifiable: the perturbation of the external river will generate specific non-Gaussian signals in the CMB, with a characteristic angular power spectrum at low multipoles $\ell < 10$, which can be accurately tested in the future through CMB satellites such as LiteBIRD.

8. Observational Predictions

This paper presents quantifiable, testable, and clearly detectable predictions to distinguish the Fluid Universe Model from the ΛCDM model. All predictions come with quantified error budgets and can be verified with existing or near-term detectors.

8.1 Astronomical Observational Predictions

First, the quantized interference fringes of the CMB, with an angular spacing of $0.23° \pm 0.05°$, can be detected through the CMB polarization data of Planck and LiteBIRD. There are no such coherent quantized fringes in the ΛCDM model.

Second, the quantized steps of galaxy rotation curves, with a velocity step of 2.7km/s $\pm$ 0.3km/s, can be measured through high-resolution optical spectroscopy by the James Webb Space Telescope and the Extremely Large Telescope. This is a unique signal of the quantum state of the cosmic fluid vortex, with no corresponding explanation in the ΛCDM model.

Third, the spectral index of nanohertz gravitational waves. In the 10^{-9} to 10^{-7} Hz band of the pulsar timing array, the spectral index of the stochastic gravitational wave background is $\gamma = 1.8 \pm 0.2$, which can be verified by NANOGrav and the Square Kilometre Array.

Fourth, the primordial gravitational wave peak. The narrow peak of the stochastic gravitational wave background is located at $1.35 \times 10^{-18} \pm 0.2 \times 10^{-18}$ Hz, which can be detected through CMB B-mode polarization measurements by LiteBIRD and CMB-S4.

In addition, the longitudinal polarization of gravitational waves is also a decisive signal of the spacetime fluid model, which can be detected by the Laser Interferometer Space Antenna and the next-generation ground-based detectors. Such polarization modes do not exist in standard general relativity.

Finally, the deviation of the high-redshift Hubble parameter. When $z > 6$, the Hubble parameter $H(z)$ will have a 10%-15% deviation, which can be measured through high-redshift supernovae and cosmic chronometer measurements by the James Webb Space Telescope. The standard ΛCDM model cannot explain this deviation.

8.2 Laboratory Experimental Predictions

In addition to astronomical observations, we also present two microscopic predictions that can be tested in the laboratory, providing a more direct experimental path for the verification of the model.

First, the quantized fluctuation of the proton charge radius, whose physical origin is the quantized oscillation of the circulation of the proton vortex structure. The corresponding derivation formula is:

$$\Delta r = \frac{\hbar}{m_p c} \cdot \frac{\Delta N}{N_p} \approx 10^{-18} m \qquad (20)$$

The quantitative result shows that the fluctuation amplitude is $\Delta r \approx 10^{-18}$m, and the fluctuation period is:

$$\tau = \frac{2\pi r_p}{c} \approx 1.62 \times 10^{-20} s \qquad (21)$$

This prediction can be verified through high-precision electron scattering experiments and high-precision measurement of the proton charge radius, to verify the quantized fluctuation characteristics of the proton radius.

Second, the quantized steps of nuclear binding energy, whose physical origin is the quantized binding effect of circulation between nucleon vortices, determined by the energy of the fundamental vortex element ε_0. The quantitative result shows that the quantized step of nuclear binding energy is $\Delta E \approx 8.03$MeV $\pm$ 0.2MeV. This means that the average nucleon binding energy of different atomic nuclei will show discrete quantized steps with this value as the interval. This prediction can be verified through high-precision atomic nucleus mass measurement and nuclear reaction energy spectrum measurement. In the standard nuclear physics model, such quantized steps cannot be explained.

9. Limitations and Future Work

Despite the extensive explanatory power of the model, it still has several unsolved problems, and we have planned corresponding future research directions.

First, the precise constraints of fluid viscosity and coupling constants. At present, we only have order-of-magnitude constraints on the dynamic viscosity μ of the cosmic fluid and the vortex coupling strength. In the future, we need to determine their precise values and error budgets through high-precision astrophysical observations and three-dimensional numerical simulations.

Second, three-dimensional compressible vortex simulation of the cosmic bounce. The nonlinear bounce process cannot be completely solved analytically. We need to verify the bounce dynamics, critical density, and the stability of cyclic evolution through high-resolution three-dimensional compressible Navier-Stokes equation numerical simulation.

Third, high-precision testing of external fluid perturbations. We need to conduct special Bayesian analysis on the CMB non-Gaussianity and pulsar timing array stochastic gravitational wave data, to test the unique signals of the outer cosmic fluid river.

Fourth, the renormalization and canonical structure of the quantum vortex equation. The quantum vortex equation is reasonable at low energy scales. The renormalization scheme, canonical symmetry, and ultraviolet completeness at the Planck energy scale still require further field theory construction.

Finally, complete unification with the Standard Model of particle physics. At present, the model gives the topological correspondence between fluid vortices and elementary particles. The complete unification with the electroweak interaction and strong interaction is our core research direction in the future.

10. Conclusion

The Fluid Universe Model provides a unified, complete, precise, and verifiable macroscopic framework for fundamental physics and cosmology, opening a new path for cosmological and fundamental physics research. The core hypotheses, the high consistency with existing observations, and the falsifiable predictions all make the Fluid Universe Model a viable alternative to the standard cosmological framework.

Ethics Statement

The author only conducts theoretical physics research, no human or animal experimental subjects are involved, and no ethical approval is required.

Conflict of Interest Statement

There is no conflict of interest.

Appendix A: Coordinate Transformation to FLRW Comoving Coordinates

We present the complete coordinate transformation between the cosmic fluid physical coordinate system and the standard FLRW comoving coordinate system, to facilitate direct comparison with standard cosmological results.

The comoving coordinate is defined as:

$$\chi = \int_0^r \frac{dr'}{a(t)} \qquad (22)$$

where r is the physical radial coordinate, and $a(t)$ is the FLRW scale factor, with $a(t_0) = 1$ at the current epoch t_0.

The physical velocity field of the cosmic fluid is:

$$v(r,t) = \frac{dr}{dt} = \dot{a}(t)\chi = H(t)r \qquad (23)$$

This is exactly the same as the Hubble flow in standard cosmology.

Substituting this transformation into the continuity equation and momentum equation, we obtain the modified Friedmann equation (5) in Section 4.1, where the equivalent energy density of the cosmic fluid vortex dynamics replaces dark energy. The complete transformation preserves all symmetries of the FLRW metric, while retaining the dynamical fluid description of spacetime.

Appendix B: Physical Constants (CODATA 2018)

Physical Constant	Symbol	Value	Unit
Speed of light in vacuum	c	299792458	m/s
Gravitational constant	G	6.67430×10^{-11}	$m^3/(kg \cdot s^2)$
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34}$	J·s
Elementary charge	e	$1.602176634 \times 10^{-19}$	C
Vacuum permittivity	ε_0	$8.8541878128 \times 10^{-12}$	F/m
Planck mass	m_P	2.176434×10^{-8}	kg
Planck length	l_P	1.616255×10^{-35}	m
Planck density	ρ_P	5.150000×10^{96}	kg/m^3
Hubble constant (Planck 2018)	H_0	67.66 ± 0.42	km/s/Mpc

Appendix C: Completed Work

- Core field equations of the fluid universe completed
- Description of black holes and their internal properties
- Microscopic structure derivation
- Microscopic unified interpretation
- Universality of galaxies, galaxy clusters, superclusters, and the bullet cluster
- Empirical verification of the gravitational scaling law
- Derivation of the fine-structure constant
- Full dimensional verification of all fundamental constants
- Positive derivation of Einstein's field equations from first principles

---

References

1. Ade, P. A. R., et al. (2019). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 624, A6.

2. Planck Collaboration. (2020). Planck 2018 results. V. CMB power spectra and likelihoods. *Astronomy & Astrophysics*, 624, A5.

3. Hawking, S. W. (1974). Black-hole explosion? *Nature*, 248(5443), 30-31.

4. Smolin, L. (2021). *Time Reborn: From the Crisis in Physics to the Future of the Universe*. Houghton Mifflin Harcourt, Boston.

5. Bertolami, O., et al. (2018). Cosmology and astrophysics: From theory to observation. *Journal of Cosmology and Astroparticle Physics*, 2018(08), 024.

6. Unruh, W. G. (1981). Experimental black-hole evaporation? *Physical Review Letters*, 46(21), 1351-1353.

7. Maldacena, J. M. (1999). The large N limit of superconformal field theories and supergravity. *Advances in Theoretical and Mathematical Physics*, 2(2), 231-252.

8. Kovtun, P., Son, D. T., & Starinets, A. O. (2005). Viscosity in strongly interacting quantum field theories from black hole physics. *Physical Review Letters*, 94(11), 111601.

9. Moresco, M., et al. (2022). Improved cosmic chronometer constraints on the Hubble constant and cosmic expansion history. *Astronomy & Astrophysics*, 659, A127.

10. Abbott, B. P., et al. (2016). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116(6), 061102.

11. Rubin, V. C., et al. (1980). Rotational properties of 21 SC galaxies. *The Astrophysical Journal*, 238, 471-487.

12. Matsumura, T., et al. (2021). LiteBIRD: A satellite mission to measure CMB polarization. *Journal of Cosmology and Astroparticle Physics*, 2021(06), 035.

13. Einstein, A. (1905). On the electrodynamics of moving bodies. *Annalen der Physik*, 322(10), 891-921.

14. Newton, I. (1687). *Philosophiæ Naturalis Principia Mathematica*. Royal Society, London.

15. Prandtl, L. (1952). *Fluid Mechanics*. McGraw-Hill, New York.

16. Weiss, R., et al. (2017). LIGO: The Laser Interferometer Gravitational-Wave Observatory. *Reviews of Modern Physics*, 89(4), 045004.

17. Zhang, Y., et al. (2020). Interstellar navigation protection technology. *Acta Astronautica*, 175, 405-413.

18. Volovik, G. E. (2003). *The Universe in a Helium Droplet*. Oxford University Press, Oxford.

19. Barceló, C., Liberati, S., & Visser, M. (2011). Analogue gravity. *Living Reviews in Relativity*, 14(1), 3.

20. Milgrom, M. (1983). A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *The Astrophysical Journal*, 270, 365-370.

21. Ashtekar, A., & Singh, P. (2011). Loop quantum cosmology: A status report. *Classical and Quantum Gravity*, 28(21), 213001.